Workshop: Two-Sample Medians

your name here

This week, we are working with the Totally Serious Pop Culture Research Institute™ to answer their research questions.

Let's call in the packages we need,

```
library(tidyverse)
library(ssstats)
```

Study 1: The Playlist Wars

The Totally Serious Pop Culture Research Institute is conducting an investigation into a question that has divided humanity for generations: Does Main Character Energy or Villain Era produce superior productivity? Researchers recruited 50 participants. Each participant completed a series of puzzles while listening to a Main Character Energy playlist. Each participant then completed another equivalent set of puzzles while listening to a Villain Era playlist. The number of puzzles completed under each playlist was recorded for every participant.

The researchers want to determine whether participants perform differently under the two playlist conditions.

```
playlist_wars <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/playlist_wars.csv")
```

```
Error:
! cannot open the connection
```

1a. Construct the appropriate q-q plot for this dataset and research question.

```
# dependent qq
playlist_wars |> dependent_qq(col1 = productivity_main_character,
                              col2 = productivity_villain)
```

```
Error:
! object 'playlist_wars' not found
```

1b. Based on your q-q plot in 1a, do we meet the necessary normality assumption? Why or why not?

No.

The normality assumption was broken because the distribution is skewed left. The qq plot also curves.

1c. Based on your answers in 1a and 1b, would you test the means or the medians here? Why?

Medians.

We are testing the medians because the normality assumption was broken.

1d. Construct the appropriate hypothesis test to determine if participants perform differently under the two playlist conditions.

Hypotheses:

- H_0 : $M_{\text{subscript}}$
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 2: The Cape Effect

The Totally Serious Pop Culture Research Institute is investigating whether superhero attire can increase confidence.

Researchers recruited 60 participants and randomly assigned each person to one of two conditions. Participants in the Cape condition were asked to wear a superhero cape while completing a short task. Participants in the No Cape condition completed the same task while wearing their normal clothing. After completing the task, participants rated their confidence on a scale from 0 to 100, with higher scores indicating greater confidence.

The researchers want to determine whether average confidence differs between participants who wore a cape and participants who did not.

```
cape_effect  <-  read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/cape_effect.csv")
```

```
Error:
! cannot open the connection
```

2a. Construct the appropriate q-q plot for this dataset and research question.

2b. Based on your q-q plot in 2a, do we meet the necessary normality assumption? Why or why not?

Your answer here.

2c. Based on your answers in 2a and 2b, would you test the means or the medians here? Why?

Your answer here.

2d. Construct the appropriate hypothesis test to determine if average confidence differs between participants who wore a cape and participants who did not.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 3: Hero vs. Villain Productivity

Researchers at the Totally Serious Pop Culture Research Institute have become concerned that people may be dramatically underestimating the effect of their fictional character identity on productivity. They recruited 80 participants and randomly assigned each participant to one of two groups. Participants assigned to the Hero group were given a short motivational prompt encouraging them to imagine themselves as the hero of an epic story. Participants assigned to the Villain group were given a prompt encouraging them to imagine themselves as the villain. Participants then had 10 minutes to complete as many puzzles as possible. The researchers recorded the number of puzzles each participant completed.

The researchers want to determine whether average productivity differs between people assigned to the Hero and Villain conditions.

```
hero_villain <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/hero_villain.csv")
```

```
Error:
! cannot open the connection
```

3a. Construct the appropriate q-q plot for this dataset and research question.

3b. Based on your q-q plot in 3a, do we meet the necessary normality assumption? Why or why not?

Your answer here.

3c. Based on your answers in 3a and 3b, would you test the means or the medians here? Why?

Your answer here.

3d. Construct the appropriate hypothesis test to determine if average productivity differs between people assigned to the Hero and Villain conditions.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 4: The Main Character Trailer Effect

The Totally Serious Pop Culture Research Institute has discovered what researchers are calling "Main Character Energy." They want to know whether an extremely dramatic movie trailer can produce it. Researchers recruited 50 participants. Before watching the trailer, each participant rated their current level of excitement on a scale from 0 to 100. Participants then watched a dramatic movie trailer containing explosions, dramatic music, slow-motion running, and at least one mysterious person saying, "Everything changes tonight." Immediately after watching the trailer, participants rated their excitement again using the same 0–100 scale.

The researchers want to determine whether participants' average excitement changed after watching the trailer.

```
trailer_effect <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/trailer_effect.csv")
```

```
Error:
! cannot open the connection
```

4a. Construct the appropriate q-q plot for this dataset and research question.

4b. Based on your q-q plot in 4a, do we meet the necessary normality assumption? Why or why not?

Your answer here.

4c. Based on your answers in 4a and 4b, would you test the means or the medians here? Why?

Your answer here.

4d. Construct the appropriate hypothesis test to determine if participants' average excitement changed after watching the trailer.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...